

Binary Search Trees

Binary Search Tree — A node-based binary tree data structure with the following properties:

e.g. A binary search tree with keys 5, 3, 7, 2, 4, 6, 8

The left subtree of a node contains only nodes with keys lesser than the node's key. The right subtree of a node contains only nodes with keys greater than the node's key.

The left and right subtree each must also be a binary search tree.

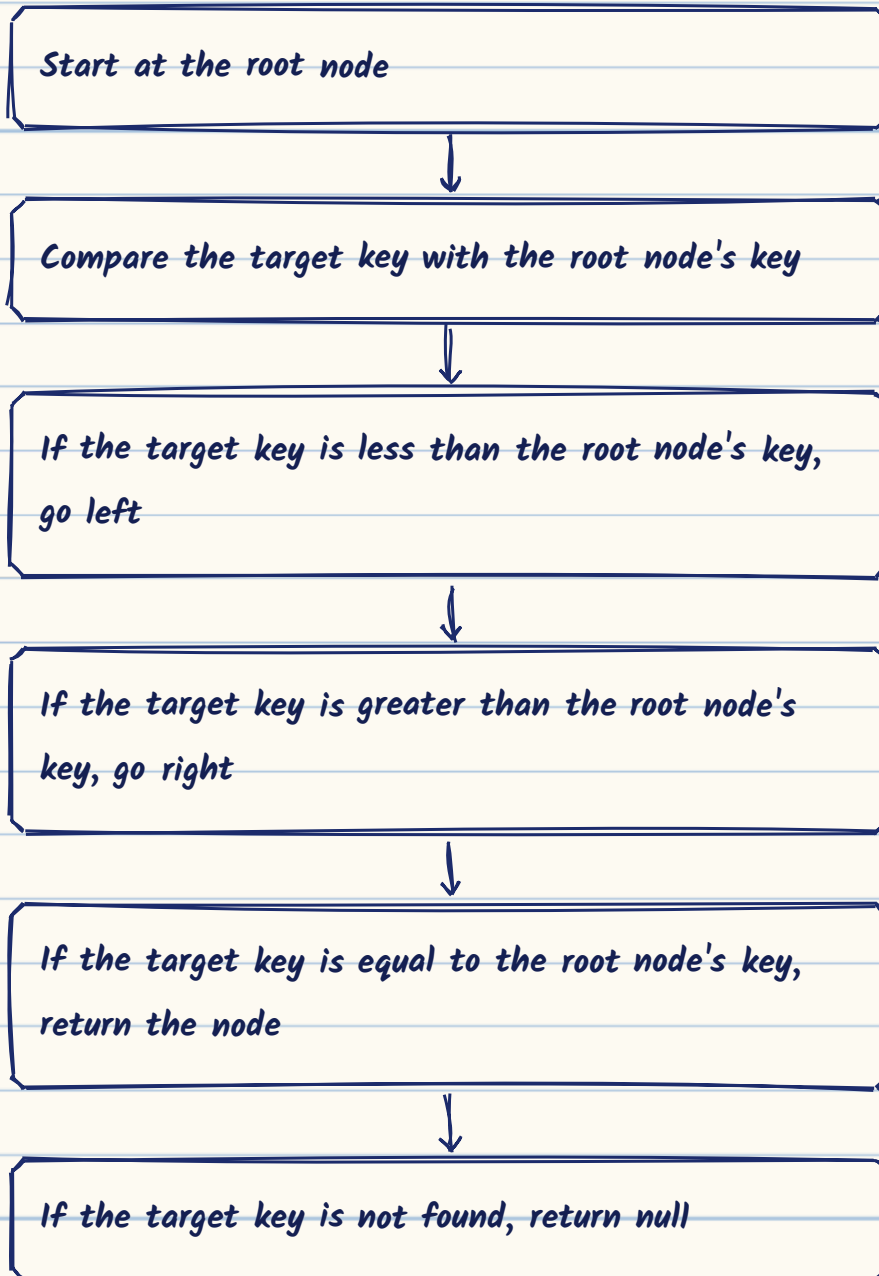
SEARCH TIME COMPLEXITY
 $O(\log n)$

Balanced BST — A binary search tree where the height of the left and right subtrees of every node differs by at most one.

e.g. A balanced binary search tree with keys 5, 3, 7, 2, 4, 6, 8

In a balanced BST, search time complexity is $O(\log n)$.

Binary Search Tree Search Algorithm



★ To maintain the BST property, insertions and deletions must be done in a way that preserves the BST property.

IN SHORT

1. A binary search tree is a node-based binary tree data structure with specific properties.
2. The left subtree of a node contains only nodes with keys lesser than the node's key.
3. The right subtree of a node contains only nodes with keys greater than the node's key.
4. The left and right subtree each must also be a binary search tree.
5. Search time complexity in a balanced BST is $O(\log n)$.